

# Lin JIU

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## EMPLOYMENT

|                 |                                                                                  |                                                           |
|-----------------|----------------------------------------------------------------------------------|-----------------------------------------------------------|
| 2023.08–Present | <b>Assistant Professor of Mathematics</b>                                        | Zu Chongzhi Center, Duke Kunshan University(DKU)          |
| 2024.07–Present | Adjunct of the Faculty of Graduate Studies                                       | Dept. of Math. & Stats., Dalhousie University             |
| 2026.07–Present | Assistant Professor of the Practice (Secondary for cross-institutional teaching) |                                                           |
| 2020.08–2025.12 |                                                                                  | Interdisciplinary Studies at DKU Studies Unit, Duke Univ. |
| 2020.08–2023.07 | Lecturer in Mathematics                                                          | Zu Chongzhi Center, DKU                                   |
| 2019.09–2020.07 | Research Associate                                                               | Dept. of Math. & Stats., Dalhousie Univ.                  |
| 2017.09–2019.08 | Killam Postdoc Fellow                                                            | Dept. of Math. & Stats., Dalhousie Univ.                  |
| 2017.03–2017.08 | Research Scientist                                                               | RICAM, Austrian Acad. Sci.                                |
| 2016.06–2017.02 | Postdoc Fellow                                                                   | RISC, Johannes Kepler Univ.                               |

## RESEARCH INTERESTS

Symbolic Computation, Number Theory, Combinatorics, Special Functions, Information Geometry

## EDUCATION

|                 |                                                          |                                   |
|-----------------|----------------------------------------------------------|-----------------------------------|
| 2011.08–2016.05 | Tulane University, Ph.D. in Mathematics                  | Advisor: <u>Victor H. Moll</u>    |
| 2013.09–2014.02 | RISC, Johannes Kepler University, Exchange Ph.D. Student | Advisor: <u>Carsten Schneider</u> |
| 2008.09–2010.07 | Beijing Institute of Technology, Master of Science       | Advisor: <u>Huafei Sun</u>        |
| 2004.09–2008.06 | Beijing Institute of Technology, Bachelor of Science     | Thesis Advisor: <u>Huafei Sun</u> |

## GRANT AWARDED

|                 |                                                                                         |                                             |               |         |
|-----------------|-----------------------------------------------------------------------------------------|---------------------------------------------|---------------|---------|
| 2023.07–2025.06 | <b>WHU-DKU Joint Grant Seed</b>                                                         | Wuhan and Duke Kunshan University           | <b>DKU PI</b> | 50K RMB |
|                 | “WHU-DKU-Dal Research Platform on Combinatorics and Number Theory”                      |                                             |               |         |
| 2023.01–2024.12 | <b>Faculty Learning Community</b>                                                       | Center for Teaching and Learning (CTL), DKU | Participant   | 10K RMB |
|                 | 11 Faculty members across disciplines   discussions on all possible teaching aspects    |                                             |               |         |
| 2022.07–2024.06 | <b>WHU-DKU Joint Grant Seed</b>                                                         | Wuhan and Duke Kunshan University           | Member        | 50K RMB |
|                 | “WHU-DKU-UMN Joint Research Center on Applied Harmonic Analysis”                        |                                             |               |         |
| 2022.01–2022.12 | <b>Gradescope Research Project Grant</b>                                                | Gradescope                                  | <b>PI</b>     | 2K USD  |
|                 | Using Gradescope in Math Courses, facilitated by CTL, DKU                               |                                             |               |         |
| 2021.07–2023.06 | <b>DKU Interdisciplinary Seed Grant</b>                                                 | DKU                                         | <b>Co-PI</b>  | 60K RMB |
|                 | “Quantum Algorithms for Computational Number Theory, Linear Algebra, and Combinatorics” |                                             |               |         |
| 2017.09–2019.08 | <b>Killam Research Fund</b>                                                             | Killam Trust @ Dalhousie University         | <b>PI</b>     | 3K CAD  |
|                 | Research Support for Killam Postdocs                                                    |                                             |               |         |

## PUBLICATIONS

(While working on the projects, undergraduate students' names are marked with underlines>

### BOOK

1. H. Sun, L. Peng, Y. Cheng, D. Li, and **L. Jiu**, *Mathematical Foundations of Information Geometry*, Science Press, Beijing, 2025. ISBN: 978-7-03-080107-4.

### PAPERS

41. **L. Jiu** and Y. Yin, Partial Results on Hankel Determinants and the Corresponding J-Fractions of a Sequence Related to Bernoulli Numbers, submitted for publication.
40. S. Chern, K. Dilcher, and **L. Jiu**, New central  $q$ -binomial identities, To Appear in *Bull. Aust. Math. Soc.*

39. **L. Jiu** and D. Wang, On  $b$ -ary binomial coefficients with negative entries, *Scientia Series A: Mathematical Sciences* **37** (2026), 69–81.
38. S. Chern, **L. Jiu**, S. Li, and L. Wang, Leading coefficient in the Hankel determinants related to binomial and  $q$ -binomial transforms, *Adv. Appl. Math.* **176** (2026), Article 103051.
37. **L. Jiu** and L. Peng, Information geometry and alpha-parallel prior of the beta-logistic distribution, *Comm. Statist. Theory Methods* **54** (2025), 3292–3306.
36. S. Chern, **L. Jiu**, and I. Simonelli, A central limit theorem for a card shuffling problem, *J. Combin. Theory Ser. A* **214** (2025), Article 106048.
35. **L. Jiu** and Y. Li, Hankel determinants of certain sequences of Bernoulli polynomials. A direct proof of an inverse matrix entry from Statistics, *Contrib. Discrete Math.* **19** (2024), 64–84.
34. Q. Chen, S. Chern, and **L. Jiu**, Multi-headed lattices and Green functions, *J. Phys. A: Math. Theor.* **57** (2024) Article 465204.
33. S. Chern and **L. Jiu**, Hankel determinants and Jacobi continued fractions for  $q$ -Euler numbers, *C. R. Math. Acad. Sci. Paris* **362** (2024), 203–216.
32. K. Dilcher and **L. Jiu**, Hankel determinants of shifted sequences of Bernoulli and Euler numbers, *Contrib. Discrete Math.* **18** (2023), 146–175.
31. Z. Bradshaw, I. Gonzalez, **L. Jiu**, V. H. Moll, and C. Vignat, Compatibility of the method of brackets with classical integration rules, *Open Math.* **21** (2023), Article 20220581.
30. **L. Jiu** and D. Y. H. Shi, Moments and cumulants on identities for Bernoulli and Euler numbers, *Math. Reports* **24** (2022), 643–650.
29. **L. Jiu**, I. Simonelli, and H. Yue, Loop decompositions of random walks and nontrivial identities of Bernoulli and Euler polynomials, *Integers* **22** (2022), A91.
28. K. Dilcher and **L. Jiu**, Hankel determinants of sequences related to Bernoulli and Euler polynomials, *Int. J. Number Theory* **18** (2022), 331–359.
27. K. Dilcher and **L. Jiu**, Orthogonal polynomials and Hankel determinants for certain Bernoulli and Euler polynomials, *J. Math. Anal. Appl.* **497** (2021), Article 124855.
26. I. Gonzales, **L. Jiu**, and V. H. Moll, An extension of the method of brackets. Part 2, *Open Math.* **18** (2020), 983–955.
25. **L. Jiu** and C. Koutschan, Calculation and properties of zonal polynomials, *Math. Comput. Sci.* **14** (2020), 623–640.
24. N. Takayama, **L. Jiu**, S. Kuriki, and Y. Zhang, Computations of the expected Euler characteristic for the largest eigenvalue of a real Wishart matrix, *J. Multivariate Anal.* **179** (2020), Article 104642.
23. **L. Jiu**, C. Vignat, and T. Wakhare, Analytic continuation for multiple zeta values using symbolic representations, *Int. J. Number Theory* **16** (2020), 579–602.
22. **L. Jiu** and C. Vignat, Connection coefficients for higher-order Bernoulli and Euler polynomials: a random walk approach, *Fibonacci Quart.* **57** (2019), 84–95.
21. **L. Jiu** and D. Y. H. Shi, Matrix representation for multiplicative nested sums, *Colloq. Math.* **158** (2019), 183–194.
20. **L. Jiu** and D. Y. H. Shi, Orthogonal polynomials and connection to generalized Motzkin numbers for higher-order Euler polynomials, *J. Number Theory* **199** (2019), 389–402.
19. I. Gonzalez, K. Kohl, **L. Jiu**, and V. H. Moll, The method of brackets in experimental mathematics, *Frontiers of Orthogonal Polynomials and  $q$ -Series*, Z. Nashed and X. Li eds., World Scientific Publishers, 2018.
18. **L. Jiu**, V. H. Moll, and C. Vignat, A symbolic approach to multiple zeta values at the negative integers, *J. Symbolic Comput.* **84** (2018), 1–13.
17. I. Gonzales, K. Kohl, **L. Jiu**, and V. H. Moll, An extension of the method of brackets. Part 1, *Open Math.* **15** (2017), 1181–1211.
16. **L. Jiu**, Integral representations of equally positive integer-indexed harmonic sums at infinity, *Research in Number Theory* **3** (2017), Article 3:10.
15. C. Li, E. Zhang, **L. Jiu**, and H. Sun, Optimal control on special Euclidean group via natural gradient descent algorithm, *Sci. China Inf. Sci.* **59** (2016), Article: 112203.
14. I. Gonzalez, **L. Jiu**, and V. H. Moll, Pochhammer symbol with negative indices. A new rule for the method of brackets, *Open Math.* **14** (2016), 681–686.
13. T. Amdeberhan, A. Dixit, X. Guan, **L. Jiu**, A. Kuznetsov, V. H. Moll, and C. Vignat, The integrals in Gradshteyn and Ryzhik. Part 30: trigonometric functions, *Scientia Series A: Mathematical Sciences* **27** (2016), 47–74.
12. T. Amdeberhan, A. Dixit, X. Guan, **L. Jiu**, V. H. Moll, and C. Vignat, A series involving Catalan numbers. Proofs and demonstrations, *Elem. Math.* **71** (2016), 109–121.
11. **L. Jiu** and C. Vignat, On binomial identities in arbitrary bases, *J. Integer Seq.* **19** (2016), Article 16.5.5.
10. **L. Jiu**, V. H. Moll, and C. Vignat, A symbolic approach to some identities for Bernoulli-Barnes polynomials, *Int. J. Number Theory* **12** (2016), 649–662.

9. A. Dixit, **L. Jiu**, V. H. Moll, and C. Vignat, The finite Fourier transform of classical polynomials, *J. Aust. Math. Soc.* **98** (2015), 145–160.
8. T. Amdeberhan, A. Dixit, X. Guan, **L. Jiu** and V. H. Moll, The unimodality of a polynomial coming from a rational integral. Back to the original proof, *J. Math. Anal. Appl.* **420** (2014), 1154–1166.
7. A. Byrnes, **L. Jiu**, V. H. Moll, and C. Vignat, Recursion rules for the hypergeometric zeta functions, *Int. J. Number Theory* **10** (2014), 1761–1782.
6. **L. Jiu**, V. H. Moll, and C. Vignat, Identities for generalized Euler polynomials, *Integral Transforms Spec. Funct.* **25** (2014), 777–789.
5. Z. Zhang, H. Sun, **L. Jiu**, and L. Peng, A natural gradient algorithm for stochastic distribution systems, *Entropy* **16** (2014), 4338–4352.
4. F. Zhang, H. Sun, **L. Jiu**, and L. Peng, The arc length variational formula on the exponential manifold, *Math. Slovaca* **63** (2013), 1101–1112.
3. L. Peng, H. Sun, and **L. Jiu**, The geometric structure of the Pareto distribution, *Bol. Asoc. Mat. Venez.* **14** (2007), 5–13.
2. **L. Jiu** and H. Sun, On minimal homothetical hypersurfaces, *Colloq. Math.* **109** (2007), 239–249.
1. X. Wang and **L. Jiu**, Characterizing hypersurfaces of generalized rotation through its normal lines, *Journal of Ningde Normal University (Natural Science)* **02** (2006), 117–119.

## ACADEMIC TALKS

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### 2026 Identities Involving Central ( $q$ -)Binomial Coefficients via ( $q$ -)Hypergeometric Series

July. 10 The SIAM Annual Meeting (AN26), Cleveland, OH, U. S. A. July 6–10.

#### Identities Involving Central ( $q$ -)Binomial Coefficients via ( $q$ -)Hypergeometric Series

Mar. 18 Invited Number Theory Seminar Talk, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.

#### Hankel Determinants of Sequences Related to Bernoulli Polynomials, Euler Polynomials, and $q$ -Series

Feb. 11 Invited Number Theory Seminar Talk, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.

### 2025 Hankel Determinants and Big $q$ -Jacobi Polynomials for $q$ -Euler Numbers

Aug. 1 The Third Joint SIAM/CAIMS Annual Meetings (AN25), Montréal, QC, Canada, July 28–Aug. 1.

#### Examples of Computer Proofs: From Elementary to Recent Ones

Jan. 15 Invited Honours Seminar Talk, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.

### 2024 Multi-headed Lattices and Green Functions

Oct. 8 Invited Number Theory Seminar Talk, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.

#### $q$ -Analogue on Hankel Determinants: the $q$ -Euler Numbers and the $q$ -Binomial Transform

June 13 Canadian Number Theory Association XVI, Fields Institute, Toronto, ON, Canada, June 10–14.

#### Shuffle to One, Shuffle to Normal

Jan. 31 Invited Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.

### 2023 Hankel Determinants on Bernoulli polynomials and $q$ -analogues

Sept. 1 Discrete Math Seminar, Zu Chongzhi Center, Duke Kunshan University, Kunshan, Suzhou, Jiangsu Province, P. R. China.

#### Random Walk Models for Identities Involving Bernoulli and Euler Polynomials

Mar. 6 Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.

### 2022 Random Walk Model on Finite Number of Sites

Oct. 19 Invited Seminar Talk, School of Mathematics, Anhui University, Online.

#### Bernoulli Symbol and Multiple Zeta Function at Non-negative Integers

Aug. 8 The First International Conference on Multiple Zeta Values and Related Topics, Online, Aug. 08–09.

#### Hankel Determinants of Certain Sequences of Bernoulli and Euler Polynomials

June 13 Invited Seminar Talk, Department of Mathematics, Zhejiang Sci-Tech University, Online.

#### Introduction on Continued Fractions, II

Mar. 25 Discrete Math Seminar, Zu Chongzhi Center, Duke Kunshan University, Kunshan, Suzhou, Jiangsu Province, P. R. China.

#### Introduction on Continued Fractions, I

Mar. 11 Discrete Math Seminar, Zu Chongzhi Center, Duke Kunshan University, Kunshan, Suzhou, Jiangsu Province, P. R. China.

#### Bernoulli and Euler Symbols: Umbral Calculus, Random Variables, and Multiple Zeta Values

Jan. 5 Duke Kunshan University-Shanghai Jiao Tong University Joint Workshop for Mathematics and Data Science, Shanghai, P. R. China.

### 2021 Random Walk Models for Non-trivial Identities Involving Bernoulli and Euler Polynomials of Higher-orders

Sept. 25 Suzhou Area Youth Mathematicians 2nd Annual Workshop, Soochow University, Kunshan, Suzhou, Jiangsu Province, P. R. China, Sept. 25–26.

#### Random Walks and Identities Involving Bernoulli and Euler Polynomials of Higher-order

June 18 Invited Seminar Talk, Institute of Statistics and Big Data, Renmin University of China, Beijing, P. R. China.

## Examples on Computer Proofs

- May 28 *Invited Seminar Talk, School of Mathematics and Statistics, Wuhan University, Wuhan, Hubei Province, P. R. China.*  
**Hankel Determinant of Sequences Related to Bernoulli and Euler Polynomials**
- May 27 *DKU-WHU Math and Stat Academic Conference, School of Mathematics and Statistics, Wuhan University, Wuhan, Hubei Province, P. R. China.*  
**2020 Hankel Determinant on Sequences Related to Bernoulli and Euler Polynomials**
- Nov. 14 *Suzhou Area Youth Mathematicians 1st Annual Workshop, Zu Chongzhi Center, Duke Kunshan University, Kunshan, Suzhou, Jiangsu Province, P. R. China.*  
**Three Examples on Computer Proofs**
- Nov. 6 *Zu Chongzhi Colloquium Series, Duke Kunshan University, Kunshan, Suzhou, P. R. China.*  
**2019 Introduction to Four Symbolic Integration Methods: Two Examples**
- Sept. 23 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**On  $b$ -ary Binomial Coefficients**
- Sept. 16 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Orthogonal Polynomials for Higher-order Euler Polynomials**
- July 23 *15th International Symposium on Orthogonal Polynomials, Special Functions and Applications, Hagenberg im Mühlkreis, Austria, July 22–26.*  
**On Harmonic Sums: Integral and Matrix Representations with Connections to Partition-theoretic Generalization of the Riemann Zeta-function and Random Walks**
- June 9 *Analytic and Combinatorial Number Theory: The Legacy of Ramanujan (A conference in honor of Bruce C. Berndt's 80th birthday), University of Illinois at Urbana-Champaign, Urbana, IL, U. S. A., June 6–9.*  
**Random Walk Approaches to Identities on Higher-order Bernoulli and Euler Polynomials**
- Mar. 17 *American Mathematical Society Spring Southeastern Sectional Meeting, Auburn University, Auburn, AL, U. S. A., Mar. 15–17.*  
**Random Walk & Identities**
- Feb. 25 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Matrix Representation for Multiplicative Nested Sums**
- Jan. 17 *2019 Joint Mathematics Meetings, Baltimore, MD, U. S. A., Jan. 16–19.*  
**Orthogonal Polynomials for Bernoulli and Euler Polynomials**
- Jan. 7 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**2018 Three Examples of Computer Proofs of Combinatorial Results**
- Oct. 17 *Honours Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Matrix Representation for Multiplicative Nested Sums**
- Sept. 21 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Bernoulli Symbol and Sum of Powers**
- July 25 *6th International Congress on Mathematical Software, University of Notre Dame, Notre Dame, IN, U. S. A., July 24–27.*  
**Random Walks and Identities for High-order Bernoulli and Euler Polynomials**
- July 3 *18th International Conference on Fibonacci Numbers and Their Applications, Dalhousie University, Halifax, NS, Canada, July 1–8.*  
**Matrix Representations for Bernoulli and Euler Polynomials**
- June 3 *2018 Canadian Mathematical Society Summer Meeting, University of New Brunswick, Fredericton, NB, Canada, June 1–4.*  
**Two Sequences Related to Bernoulli and Euler Numbers**
- May 30 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Hidden Walks**
- Feb. 26 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Introduction to Zonal Polynomials**
- Jan. 22 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**2017 The Probabilistic and Combinatorial Interpretations of the Bernoulli Symbol**
- Dec. 10 *2017 Canadian Mathematical Society Winter Meeting, University of Waterloo, Waterloo, ON, Canada, Dec. 8–11.*  
**Bernoulli Symbol  $\mathcal{B}$ : from Umbral Calculus to Random Variable and Combinatorics**
- Oct. 13 *Number Theory Seminar, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Visualization of Bernoulli Numbers**
- Oct. 12 *Colloquium, Department of Mathematics and Statistics, Dalhousie University, Halifax, NS, Canada.*  
**Bernoulli Symbol on Multiple Zeta Values at Negative Integers**
- July 18 *23rd Conference on Applications of Computer Algebra (Commemorating the heritage of Jonathan Michael Borwein), Jerusalem College of Technology, Jerusalem, Israel, July 17–21.*  
**On Bernoulli Symbol  $\mathcal{B}$**
- May 5 *Klagenfurt-Linz-Wien Workshop, Riefnitz, Austria, May 3–6.*  
**2016 The Method of Brackets (MoB) and Integrating by Differentiating (IBD) Method**
- Dec. 9 *Laboratoire des Signaux et Systèmes, Université Paris Sud XI, Orsay, France.*  
**“Random Walks” for Harmonic Sums**
- Nov. 29 *SFB Statusseminar, Strobl, Austria, Nov. 27–30.*

**A Hot Pot**

Oct. 5 Algorithmic Combinatorics Seminar, Research Institute for Symbolic Computations, Johannes Kepler University, Hagenberg im Mühlkreis, Austria.

**On Binomial Identities in Arbitrary Bases**

July 26 Beijing Key Laboratory on Mathematical Characterization, Analysis and Applications of Complex Information, Beijing Institute of Technology, Beijing, P. R. China.

**Random Walk: A Probabilistic and Geometric Approach to Number Theory**

July 19 International Conference on Mathematical Characterization, Analysis and Applications of Complex Information, Beijing Institute of Technology, Beijing, P. R. China, July 19–20.

**The Method of Brackets**

July 13 5th International Congress on Mathematical Software, The Zuse Institute Berlin, Berlin, Germany, July 11–14.

**The Method of Brackets**

June 22 Algorithmic Combinatorics Seminar, Research Institute for Symbolic Computations, Johannes Kepler University, Hagenberg im Mühlkreis, Austria.

**Binomial Identities in Arbitrary Bases**

Mar. 8 Graduate Students Colloquium, Department of Mathematics, Tulane University, New Orleans, LA, U. S. A.

**2015 On Bernoulli Symbol  $\mathcal{B}$  and Its Applications**

July 8 Invited Seminar Talk, Center for Combinatorics, Nankai University, Tianjin, P. R. China

**2014 Recursion Rules for the Hypergeometric Zeta Functions**

June 3 Midwest Number Theory Conference for Graduate Students and Recent PhDs, X, University of Illinois at Urbana-Champaign, Urbana, IL, U. S. A., June 3–4.

**Implementation of an Algorithm on Converting Sums into Nested Sums**

Jan. 8 Invited Seminar Talk, Laboratoire des Signaux et Systemes, Université Paris Sud XI, Orsay, France.

**HONORS AND AWARDS**

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|                                                                                       |                                      |
|---------------------------------------------------------------------------------------|--------------------------------------|
| 2016 Tea Doctor (for organizing Tea Time)                                             | Dept. of Math., Tulane Univ.         |
| 2015 Tea Master (for organizing Tea Time)                                             | Dept. of Math., Tulane Univ.         |
| 2014 Excellence in Mathematics                                                        | Dept. of Math., Tulane Univ.         |
| 2013 Excellent Graduate Student Teacher                                               | Dept. of Math., Tulane Univ.         |
| 2008 Outstanding Graduates                                                            | Beijing Institute of Technology      |
| 2007 National Scholarship                                                             | Department of Education, P. R. China |
| 2006 China Aerospace Science and Technology Corporation Scholarship (CASC), 2nd Prize | CASC, P. R. China                    |

**TEACHING EXPERIENCE**

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**TEACHING SUMMARY** 29 regular course sections (incl. 5 scheduled), 1 mini-term, and 6 independent study courses across **Duke Kunshan Univ. (DKU)**, **Dalhousie Univ. (Dal)**, and **Tulane Univ. (TU)**.

|     |                                                             |                                                       |
|-----|-------------------------------------------------------------|-------------------------------------------------------|
| DKU | MATH101A—Introductory Calculus I: Functions and Derivatives | 2026 Fall (scheduled)                                 |
|     | MATH101B—Introductory Calculus II: Integration              | 2027 Spring (scheduled)                               |
|     | MATH105—Calculus                                            | 2020–2023 ( <b>5</b> sections)                        |
|     | MATH201—Multivariable Calculus                              | 2026 Fall (scheduled), 2020–2022 ( <b>3</b> sections) |
|     | MATH202—Linear Algebra                                      | 2026 Fall (scheduled), 2023 Fall                      |
|     | MATH205—Probability and Statistics                          | 2023 Spring, 2021 Spring                              |
|     | MATH301—Advanced Introduction to Probability                | 2021–2023 ( <b>4</b> sections)                        |
|     | MATH306—Number Theory                                       | 2022 Fall, 2021 Fall                                  |
|     | MATH307—Complex Analysis                                    | 2027 Spring (scheduled), 2024 Fall                    |
| Dal | MATH1030—Matrix Theory and Linear Algebra I                 | 2019 Summer                                           |
|     | MATH3080—Complex Variables                                  | 2019 Winter                                           |
| TU  | MATH1160—Long Calculus II                                   | 2014 Summer, 2016 Spring                              |
|     | MATH1210—Calculus I                                         | 2015 Spring                                           |
|     | MATH1310—Consolidated Calculus                              | 2015 Fall                                             |

### Mini-term & Reading/Independent Study Courses

|     |                                                                                           |                                                               |
|-----|-------------------------------------------------------------------------------------------|---------------------------------------------------------------|
| Dal | MATH6200                                                                                  | 2025 Winter—Integer Partitions and $q$ -Series                |
| DKU | MINITERM102                                                                               | 2023 Spring—Experimental Mathematics and Symbolic Computation |
|     | INDSTU391                                                                                 | 2021–2026 ( <b>5</b> sections)                                |
|     | 2026 Fall (scheduled) —Introduction to Advanced Probability                               |                                                               |
|     | 2022 Fall—Introduction to Algebraic Geometry, 2022 Spring—Variational Quantum Algorithms, |                                                               |
|     | 2021 Fall, 3 sections—Riemann Zeta-Function, Quantum Algorithm, Combinatorics             |                                                               |

## ACADEMIC SERVICES AND MEMBERSHIPS

|              |                                                                                                                                                                                                   |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026         | Organizer of the mini-symposium <i>Special Functions with Applications in Number Theory and Combinatorics</i> at The SIAM Annual Meetings, July 6–10, Cleveland, OH, U.S.A.                       |
| 2025–Present | Reviewer for Mathematical Reviews @ AMS                                                                                                                                                           |
| 2025         | Organizer of the mini-symposium <i>Special Functions with Applications in Number Theory and Combinatorics</i> at The Third Joint SIAM/CAIMS Annual Meetings, July 27—Aug. 1, Montréal, QC, Canada |
| 2014–Present | Reviewer for journals including: Journal of Number Theory, The Ramanujan Journal, Journal of Difference Equations and Applications, etc.                                                          |

## DEPARTMENTAL AND UNIVERSITY SERVICE

|              |                                                                  |                         |
|--------------|------------------------------------------------------------------|-------------------------|
| 2021–Present | Organizer of the Discrete Math Seminar                           | Duke Kunshan University |
| 2024         | Member of 2024 Undergraduate Recruitment & Admissions Evaluation | Duke Kunshan University |
| 2017–2020    | Organizer of the Number Theory Seminar                           | Dalhousie University    |
| 2012–2016    | Organizer of the Tee Time                                        | Tulane University       |

## STUDENT MENTORING, ADVISING, AND RESEARCH ACTIVITIES

### Undergraduate Academic Advisor @ DKU

- Served as primary academic advisor for **25** students (Classes of 2022–2029).
- Provide continuous guidance on course selection, career planning, and graduate school applications.

### Undergraduate Signature Work (SW)<sup>1</sup> ( $\cong$ Honor Thesis) Mentor @ DKU & Undergraduate Student Research Projects

- Mentored **10 SW projects** (Class of 2023–24) @DKU on topics ranging from Random Walks to Quantum Algorithms.
- Undergraduate Student Research Highlights & Outcomes:
  - Awarded Summer Research Program<sup>2</sup> for the years 2022, 2023, and 2026.

<sup>1</sup><https://signature-work.dukekunshan.edu.cn/signature-work-overview/>

<sup>2</sup><https://ugstudies.dukekunshan.edu.cn/student-summer-research-program/>

- Peer-Reviewed Publications: Guided 4 students to co-author papers in international journals (see items [29], [35], [38], [39] in Publications/Papers).
- Software Development: Directed the development of 2 Mathematica packages for Method of Brackets and Weakly Increasing Trees, leading to SW projects.
- International Collaboration: Co-mentored international research on Cyclotomic Polynomials with Prof. K. Dilcher (Dalhousie University).

## RELEVANT SKILLS

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*Language:* Mandarin (native), English (fluent)

*Computer:* Mathematica, SageMath, Python, Maple,  $\text{\LaTeX}$ , LyX

*Packages:* Zonal.sage <https://jiulin90.github.io/Packages/Zonal.sage>

BNE.sage <https://jiulin90.github.io/Packages/BNE.sage>